Captions give words. Not signing.

Who feels it

Deaf and hard-of-hearing viewers of short Bangla clips, and the creators who publish them. (hypothesis, not yet validated)

BdSL is not ASL

Bangla Sign Language needs its own signs. Generic tools do not help.

Silent failure

Automated signing can drop a word with no warning. “**দুধ নেই**” (there is no milk) becomes “milk”.

What started this

In many countries, ordinary people learn sign language to help deaf people. In Bangladesh that practice barely exists, so deaf people are left out of everyday conversation. That gap is why I built this.

A signing companion that shows its gaps.

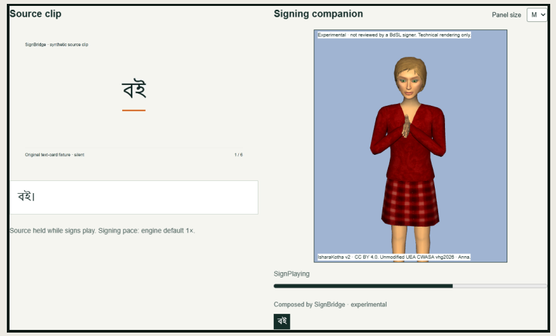
Upload a short clip plus its Bangla captions.

Build a saved, source-timed signing track from a real BdSL corpus.

Replay: the video holds while the avatar signs each segment.

Gaps stay visible before signing, never silently dropped.

Every composition is labelled: Experimental · not reviewed by a BdSL signer.



Source held while the avatar signs



Unsupported negation: a blocking gap

Model proposes. Code and SQL decide.

Segment	Meaning	Retrieve	Plan	Validate	Publish
rules	AI	code	AI	code	SQL

Constrained AI

The plan can only pick sign IDs from retrieved candidates. No invented signs. An independent detector guards negation, numbers and questions.

SQL is the authority

Job and epoch checks fence stale work. Up to 3 tries, then one repair. Spend caps and a kill switch.

Two clocks

Source and avatar run separately. Only ordered unit completion counts as signed.

Next.js · Vercel Workflows · Neon Postgres · OpenAI Responses (store:false) · UEA CWASA avatar · IsharaKotha corpus (CC BY 4.0)

Browser → API → Workflow → SQL / Blob / model / avatar iframe

Solo builder.

Participant

Md Tayeb Ibne Sayed
Solo entry · YUNet AIgnite primary round

`github.com/Tayebbb/signbridge`

How it was built

One person, with AI coding tools. Designed around evidence: every claim in this deck traces to a test, a capture or a recorded limit.

Why this problem

I saw that abroad many hearing people learn sign language to help deaf people. In Bangladesh almost no one does. I wanted to understand why this bothers deaf people, and to build something that helps.

Try it

Two public, read-only synthetic samples at

`signbridge-mdtayebibne-4755s-projects.vercel.app`

What is proven. What is not.

Technical: yes

194 tests · 51 native PostgreSQL · 16 browser. 20/20 sampled and 12/12 curated assets render. 6 production segments replayed in Chrome and Edge. USD 0.10 spent.

Linguistic: none

No BdSL signer has reviewed these renders.

Comprehension: none

No study with Deaf / BdSL users yet.

Next

Co-designed review with qualified signers → comprehension tasks → real Android and iPhone testing → wider vocabulary, no silent substitution.

Not a first. The tested contribution is the combination: source-frame holds, durable caption edits, saved replay and visible gaps.

Prior art credited: SignOLight, IsharaKotha